

Swetha Ashok

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Education

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering and Computer Science, Minor in Energy Engineering

GPA: 3.5 • Expected Graduation May 2028

Relevant Coursework: Python, Java & Data Structures, Circuits & Devices, PCB Design, C & Computer Architecture

Expected Completion Spring 2026: Microfabrication Technology, Microelectronic Devices

Skills & Interests

Hardware: PCB Design, soldering, subsystem integration, debugging, oscilloscopes, digital multimeters

Software: Python, Java, C, RISC-V, Github, MS Suite, LaTeX

CAD: SOLIDWORKS, Onshape, Autodesk Fusion 360, ANSYS (FEA), KiCAD

Languages: English (Native), Tamil (Native), Spanish (Conversational)

Technical Experience

Calsol – Solar Racing Team

Electrical Member, Solar Subteam Advisor (Current), Solar Project Manager

August 2024 – PRESENT

- Led a 20+ member team in training for solar module wiring, bypass diode soldering, and PV cell integration.
- Designed integration of the solar array into battery via MPPT, coordinating with electrical and mechanical teams on cross-functional requirements leading to expedited timelines.
- Spearheaded design and hardware build of a mini solar car project, using repurposed PV cells and material.

Retro Synth

Co-Designer and Developer

January 2026 – December 2026

- Delivered a fully functional 12-note analog PCB synthesizer on a 12-week development timeline by collaborating in a 3-person engineering team across design, hardware bring up and custom firmware implementation.
- Designing schematics using **KiCAD** and managing BOM to meet performance and budget constraints.

FIRST Robotics Competition Team 2643

President • Project Manager • Safety Captain

August 2020 – May 2024

- Directed the design, wiring, and testing of multi-sensor robotic systems using **SOLIDWORKS**, enabling the 50-member team to build functional prototypes and competition ready robots that met electrical and mechanical specifications.
- Streamlined subsystem assembly through the master CAD and driving synchronous hardware/software milestones.

Leadership Experience

Society of Women Engineers

Executive, Treasurer (Current) • Academic Chair • Advocacy Chair

August 2024 – PRESENT

- Accelerated reimbursement turnarounds by 75% across 100+ active members by managing budget allocations and financial counseling for 30+ distinct programs.
- Preparing 25 collegiate attendees for industry recruitment by organizing logistics and career-prep workshops for WE National and Regional Conferences.

UC Berkeley College of Engineering, Student Services

Peer Advisor

August 2026 – May 2027

- Advise prospective and undergraduate engineering students on college requirements, major options, and administrative policies such as course planning and major changes.
- Deliver peer-to-peer guidance on academic strategies, course workload balance, and navigating campus resources, research, internships, and student organizations.
- Support the Engineering Advisors with outreach events, informational sessions and workshops throughout the semester to support students' academic transition and holistic development within Berkeley Engineering.